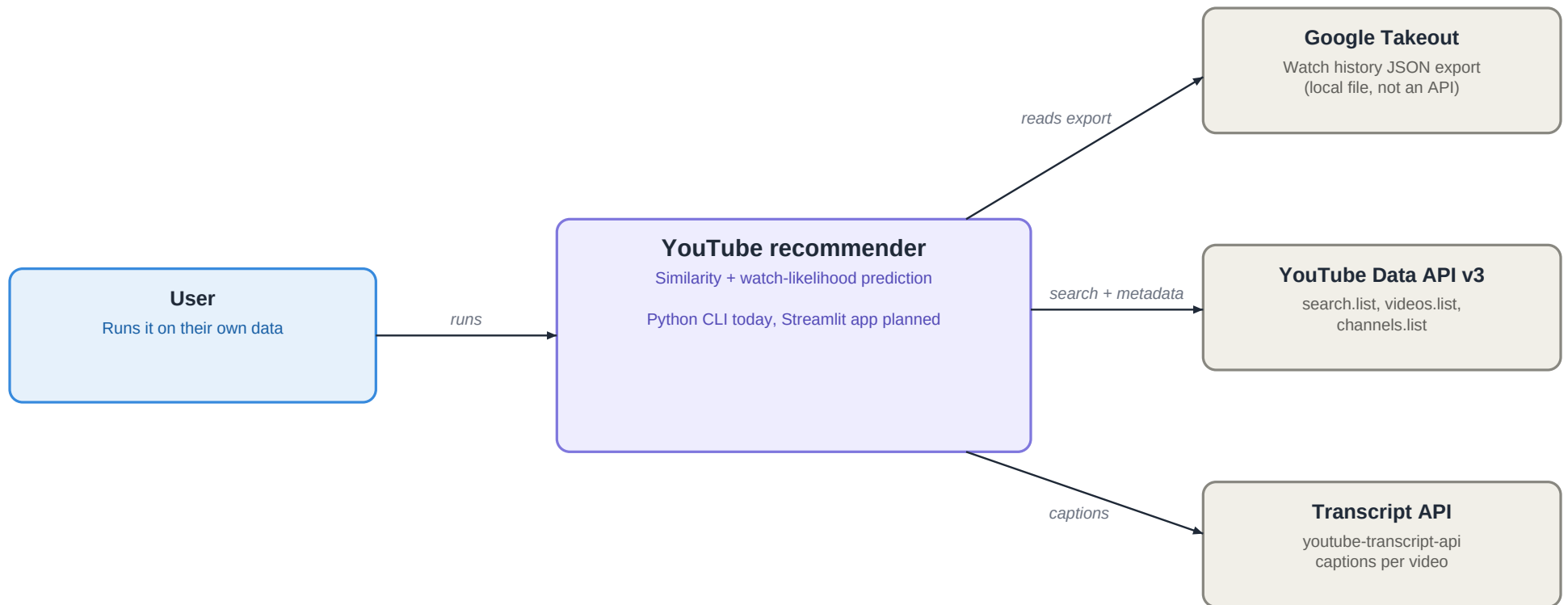


Level 1: system context

Who uses the system and what it depends on



The user supplies their own Takeout export, so no one else's watch history is ever pulled.

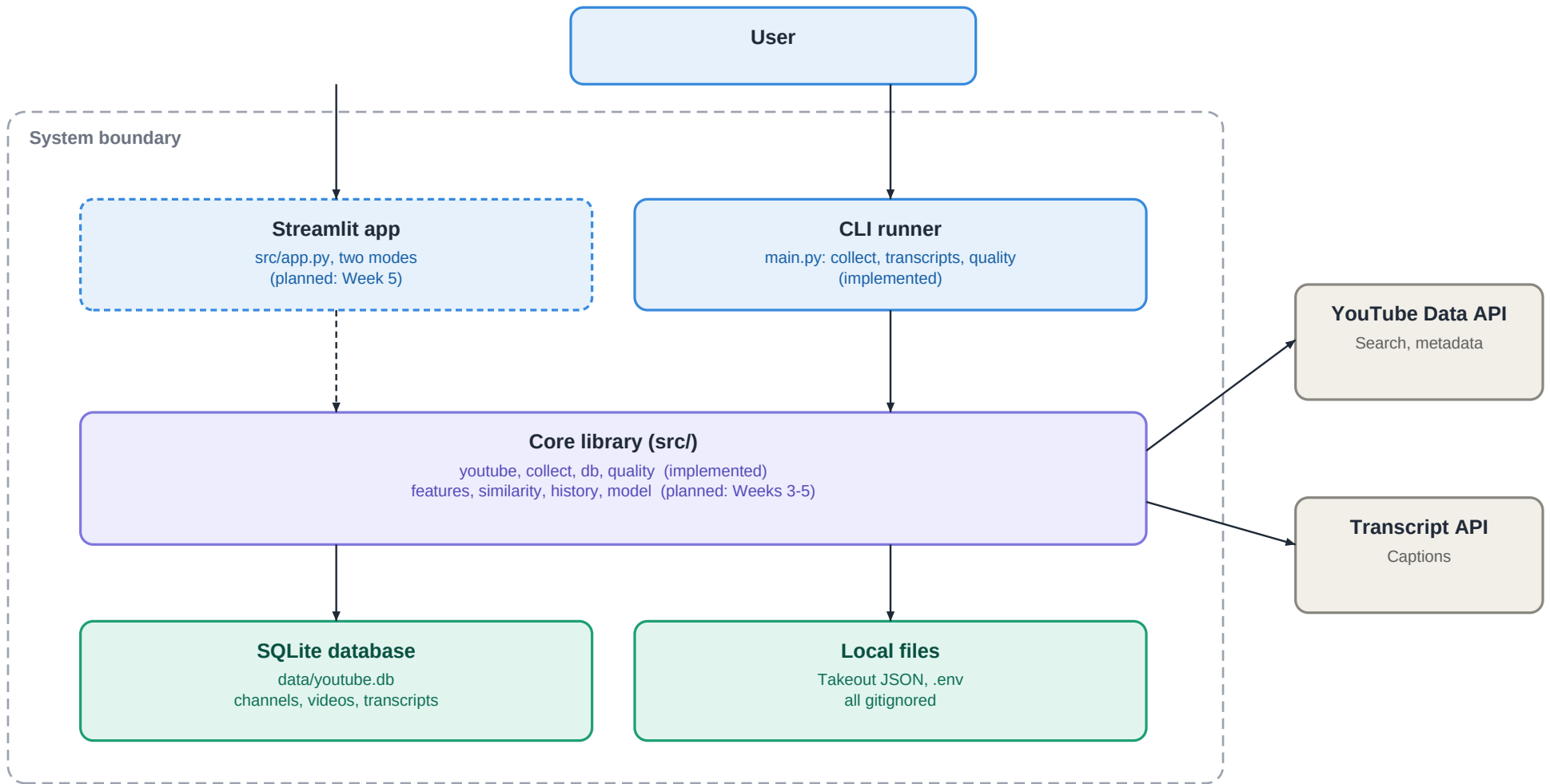
The YouTube Data API provides search results and full video metadata (duration, stats, tags). search.list costs 100 quota units per call.

Transcripts come from a separate captions library; 10-30% of videos may have none, and the pipeline records that instead of failing.

 Person This system External system / file

Level 2: containers

What runs inside the system boundary



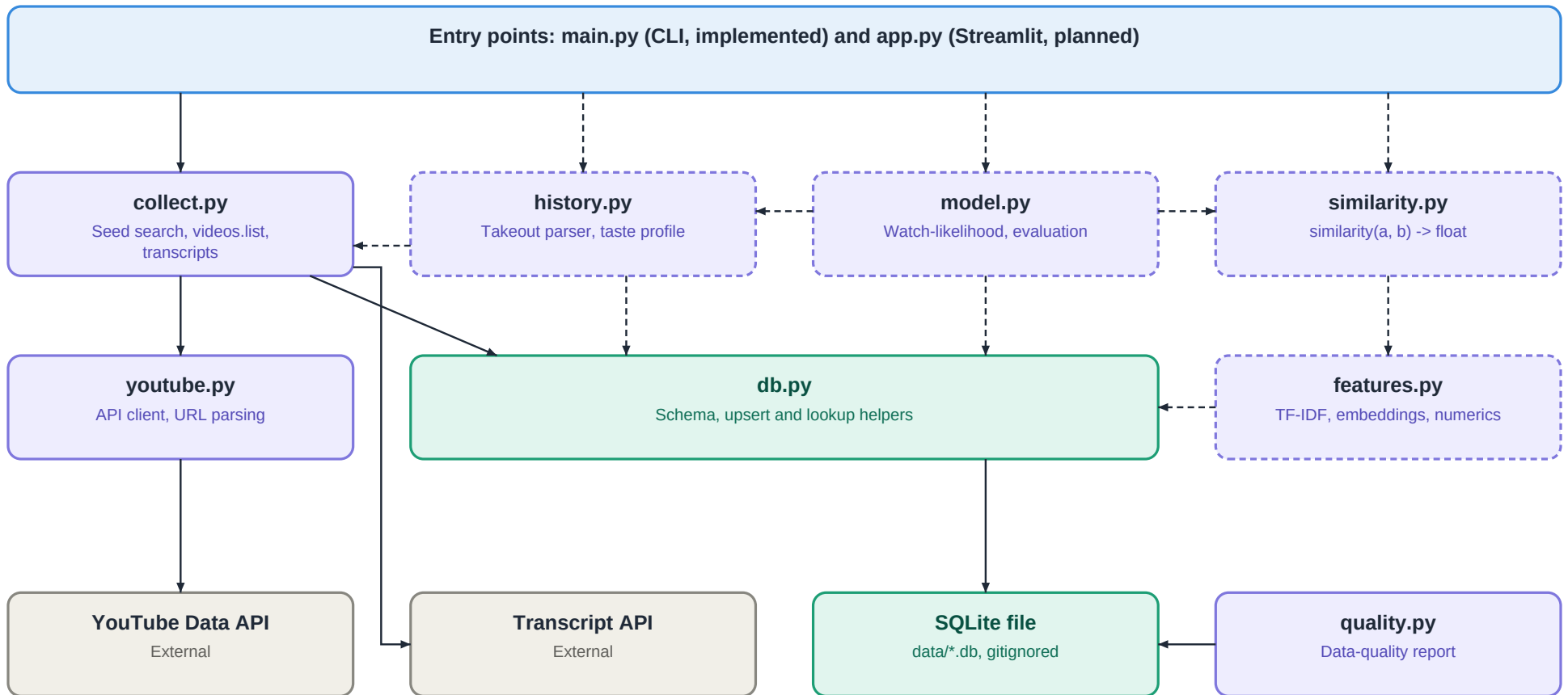
Both entry points call the same core library; only the core touches the database, local files and external APIs.

quality.py reads the database through the connection main.py opens; collect.py calls the Transcript API directly and the YouTube API via youtube.py.

 Entry point Code container Data store External Dashed outline = planned

Level 3: components of the core library

Modules in src/ and how they depend on each other

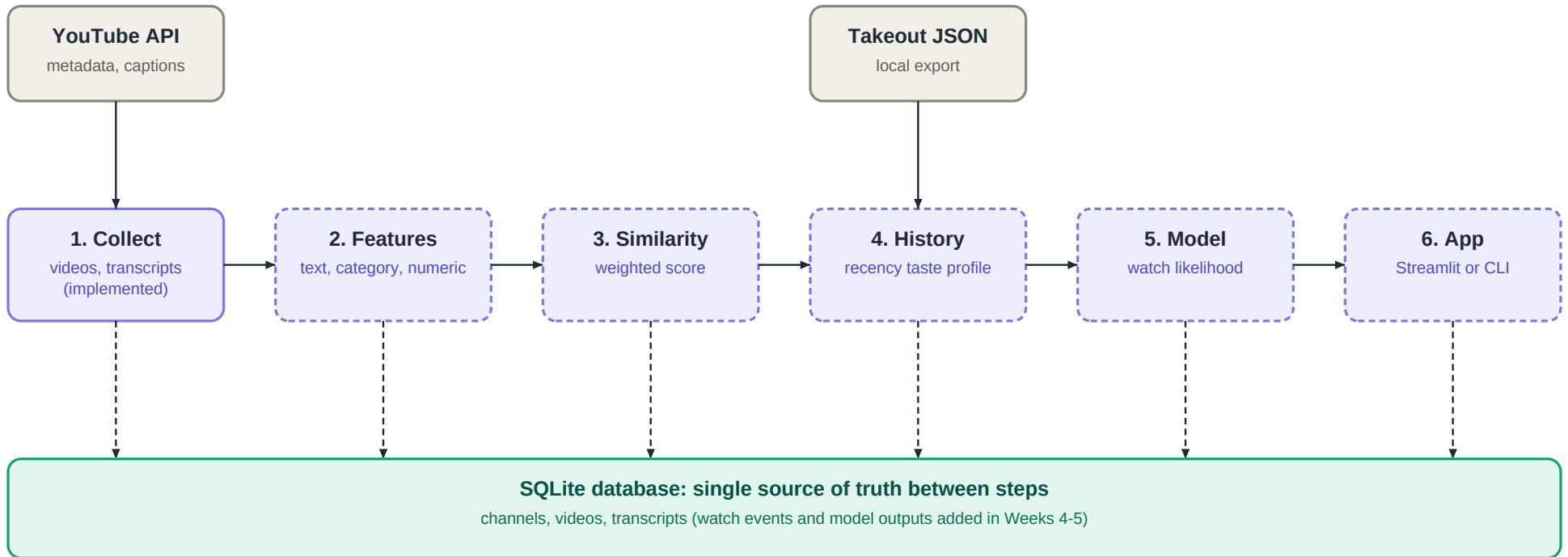


model.py builds on history.py (taste profile) and similarity.py (scores); similarity.py builds on features.py.
history.py enriches watched videos through collect.py, and reads the user's Takeout JSON from local files.
db.py owns the schema and write helpers; collect.py and quality.py also run read queries on the same connection.
youtube.py wraps the YouTube Data API; collect.py calls the Transcript API (youtube-transcript-api) directly.

Entry points Module Data access / store External Dashed = planned

Runtime data flow

The pipeline from collection to recommendations



Dashed lines show reads from and writes to the database, so iterating on features never re-spends API quota.

Quota check: search.list costs 100 units per call, so 6 queries x 2 pages = 1,200 of the 10,000 daily units; metadata calls cost 1 unit per 50 videos.

Labels for the model are watched (1) vs. not watched (0), including hard negatives, with a time-based train/test split.

Evaluation uses precision, recall, F1 and ROC-AUC rather than accuracy, because unwatched videos vastly outnumber watched ones.

Pipeline stage Data store External input Dashed = planned